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Method and device for uterine fibroid treatment

Abstract

Methods and devices for both imaging and treating uterine fibroid tumors in one real-time system are provided. One minimally invasive method comprises introducing a sheath into a uterus and determining a location of a fibroid using a visualization element within or on the sheath. Upon identification, a portion of the sheath is steered to position at least one treatment needle at the determined location. The needle is then anchored in uterine tissue and the fibroid treated with the needle.

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Background/Summary

CROSS-REFERENCE (1) The present application is a continuation of U.S. patent application Ser. No. 17/863,988, filed Jul. 13, 2022, which is a continuation of U.S. patent application Ser. No. 16/221,138, filed Dec. 14, 2018, now U.S. Pat. No. 11,419,668, which is a continuation of U.S. patent application Ser. No. 15/970,428, filed May 3, 2018, now U.S. Pat. No. 10,182,862; which is a continuation of U.S. patent application Ser. No. 15/720,199, filed Sep. 29, 2017, now U.S. Pat. No. 9,987,080; which is a continuation of U.S. patent application Ser. No. 12/973,642 filed Dec. 20, 2010, now U.S. Pat. No. 9,808,310; which is a continuation of U.S. patent application Ser. No. 11/347,018 filed Feb. 2, 2006, now U.S. Pat. No. 7,918,795; which claims the benefit of Provisional Patent Application Nos. 60/710,712, filed Aug. 22, 2005, and 60/649,839, filed Feb. 2, 2005; the full disclosures of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates generally to medical devices and methods. More particularly, the invention relates to methods and devices for locating and treating uterine fibroids.

(2) There are unmet needs in the pathophysiology of the female reproductive tract, such as dysfunctional uterine bleeding and fibroids. Fibroids are benign tumors of the uterine myometria (i.e., muscle) and are the most common tumor of the female pelvis. Fibroid tumors affect up to 30% of women of childbearing age and can cause significant symptoms such as discomfort, pelvic pain, menorrhagia, pressure, anemia, compression, infertility and miscarriage. Fibroids may be located in the myometrium (i.e., intramural), adjacent to the endometrium (i.e., submucosal), or in the outer layer of the uterus (i.e., subserosal). Most commonly fibroids are a smooth muscle overgrowth that arise intramurally and can grow to be several centimeters in diameter.

(3) Current treatment for fibroids includes medical treatment with NSAIDS, estrogen-progesterone combinations, and GnRH analogues. Pharmacologic therapy with GnRH analogues is limited due to its side effects, such as hot flashes, vaginal dryness, mood changes and bone density loss. Further, its relatively short time of treatment (e.g., 3 months) offers temporary shrinkage, wherein the fibroids usually regrow after medical discontinuation. Pharmacologic therapy is relatively ineffective and palliative rather than curative.

(4) Hysterectomy (i.e., surgical removal of the uterus) is a common treatment for fibroids. It is performed up to 600,000 times annually in the United States. Indeed, fibroids are the indication for hysterectomy in up to one third of all cases. Hysterectomy for treating fibroids may be very effective but has many undesirable side effects such as loss of fertility, open surgery, sexual dysfunction, and long recovery time. There is also significant morbidity (e.g., sepsis, hemorrhage, peritonitis, bowel and bladder injury), mortality, and costs associated with hysterectomy treatments.

(5) Surgical myomectomy is also an open surgical procedure requiring laparotomy and general anesthesia in which fibroids are removed. Often these procedures result in significant blood loss and can only remove a portion of the culprit tissue. In the early 1990's, there was a growth in advanced operative laparoscopy techniques and laparoscopic myomectomy was pioneered. However, laparoscopic myomectomy remains technically challenging. For example, it requires

laparoscopic suturing which is performed only by the most skilled of laparoscopic gynecologists. Further, prolonged anesthesia time, increased blood loss, and possibly higher risk of uterine rupture in pregnancy make laparoscopic myomectomy a challenging procedure. Currently, the removal of subserosal or intramural fibroids requires an abdominal approach.

(6) Hysteroscopy (i.e., process by which a thin fiber optic camera is used to image inside the uterus) may include an attachment to destroy tissue. Hysteroscopic resection is a surgical technique that uses a variety of devices (e.g., loops, roller balls, bipolar electrodes) to ablate or resect uterine tissue. The uterus needs to be filled with fluid for better viewing and thus has potential side effects of fluid overload. Hysteroscopic ablation is also limited by its visualization technique and is thus only appropriate for those fibroids that are submucosal and/or protrude into the uterine cavity.

(7) Uterine artery embolization has also been suggested as an alternative minimally invasive treatment for fibroids. Uterine artery embolization was introduced in the early 1990's and is performed by injecting small particles through a groin incision into the uterine artery to selectively block the blood supply to fibroids. Uterine artery embolization results in reduction of the myoma size from 20-70% at six months. However, side effects of this procedure include pelvic infection, premature menopause, and severe pelvic pain. In addition, long term MRI data suggest that incomplete fibroid infarction may result in regrowth of infarcted fibroid tissue.

(8) Despite much interest in uterine artery embolization, the procedure rates remain low and have not grown past a few thousand performed per year in the United States. This may be due to the fact that interventional radiologists, instead of gynecologists who know how to diagnose and treat fibroid tumors, are the ones who perform uterine artery embolization procedures.

(9) Endometrial ablation, which is primarily a procedure for dysfunctional or abnormal uterine bleeding, may be used at times for fibroids. Recently there have been many new technologies to perform endometrial ablation such as cryo energy, microwave energy, and impedance controlled radiofrequency. Endometrial ablation destroys the endometrial tissue lining the uterus, but does not specifically treat fibroids. This technique is also not suitable for women who desire to bear children. Endometrial ablation remains a successful therapy for dysfunctional uterine bleeding, but is limited in its ability to treat fibroids.

(10) Myolysis is another alternative minimally invasive technique for fibroid treatment. Myolysis was first performed in the 1980's in Europe by using lasers to coagulate tissue, denature proteins, and necrose myometrium with laparoscopic visualization. This technique has been in use for the past several years and involves applying energy directly to the myoma. Laparoscopic myolysis can be an alternative to myomectomy, as the fibroids are coagulated and then undergo coagulative necrosis resulting in a dramatic decrease in size. Shrinkage of fibroids has been reported at 30-50%. In addition, there is the obvious economic benefit of out-patient surgery, rapid recovery, and return to normal lifestyle. However, all laparoscopic techniques are limited by the fact that they can only see, and therefore only treat, subserosal fibroids.

(11) Needle myolysis is a promising technique whereby a laparoscope is used to introduce one or more needles into a fibroid tumor under visual control. Bipolar radiofrequency current is then delivered between two adjacent needles, or monopolar current between a single needle and a distant dispersive electrode affixed to the thigh or back. The aim of needle myolysis is to coagulate a significant volume of the tumor and thereby cause it to shrink substantially. The traditional technique is to make multiple passes through different areas of the tumor using the coagulating needle to destroy many cylindrical cores of abnormal tissue. However, the desirability of multiple passes is mitigated by the risk of adhesion formation, which is thought to increase with increasing amounts of injured uterine serosa and by the operative time and skill required.

(12) For these and other reasons, it would be desirable to provide a minimally invasive method and device to selectively eradicate fibroid tumors within the uterus. It would be desirable if the method and device could locate and treat all types of fibroids in the uterus in a safe and effective manner with minimum risk and discomfort for the patient. It would be further desirable to provide a

method and device for eradicating fibroid tumors that combines imaging and treatment in one simple hand held device. At least some of these objectives will be met by the methods and devices of the present invention described hereinafter.

Description of the Background Art

(13) Relevant references include 5,456,689 Kresch et al.; U.S. Pat. No. 5,979,453 Savage et al.; U.S. Pat. No. 6,002,968 Edwards; U.S. Pat. No. 6,550,482 Burbank et al.; U.S. Pat. No. 6,626,855 Weng et al.; U.S. Pat. No. 6,716,184 Vaezy et al.; WO 2004/064658; and US 2005/0107781. The full disclosures of each of the above references are incorporated herein by reference.

SUMMARY OF THE INVENTION

(14) In a first aspect of the present invention, methods for minimally invasive identification and treatment of submucosal, intramural, or subserosal fibroids of the uterus are provided. A sheath, catheter, or probe may be transcervically introduced into the uterus. A location of the fibroid tumor may be determined by using a visualization element within or on the sheath. Preferably, the physician will be able to image the tumors transendometrially from within the uterine cavity. The visualization element may comprise an ultrasonic element or other visualization means, such as hysteroscopy, that is capable of producing a visual image. Once having identified the location, a portion of the sheath is steered to position at least one treatment needle at the determined location. The needle is anchored within the uterine tissue and the fibroid is treated with the needle. Fibroid treatment may take several forms as discussed in more detail below. Generally, each individual fibroid tumor will be navigated to, imaged, targeted and treated separately. It will further be appreciated that external imaging may be preformed prior to sheath introduction so as to initially “map” the location of the fibroid tumors.

(15) Anchoring comprises manually positioning and penetrating the treatment needle through an endometrium so as to engage the fibroid. Preferably, the visualization element not only provides a field of view for locating the fibroid but also provides a field of view for directly observing and verifying needle deployment and fibroid treatment in real-time. Visualization may be aided by steering, rotating, or deflecting the visualization element independent of the treatment needle so as to provide a complete view. At least one treatment needle, preferably two treatment needles, will be anchored in the uterine tissue so that the needles will remain substantially immobile during the delivery of treatment. For example, anchoring may comprise deploying at least two treatment needles in a converging manner so as to pinch the fibroid therebetween. Alternatively, anchoring may comprise deploying at least two treatment needles in a diverging manner so as to hook the fibroid therebetween. Still further, anchoring may comprise deploying at least two treatment needles in a telescoping manner.

(16) The uterine fibroid may be treated in several ways. Usually the fibroid will be treated by delivering ablative energy to the fibroid with the needle to necrose the tissue. The ablative energy may comprise electrically energy (e.g., radiofrequency energy, laser energy, or microwave energy), freezing energy (e.g., cryo energy), ultrasound energy, high intensity focused ultrasound (HIFU), or radiation. Preferably, the treatment needle comprises electrically conductive electrodes that deliver ablative radiofrequency energy in a bipolar or monopolar fashion. In addition to or in lieu of ablative energy treatment, the fibroids may be treated by delivering at least one therapeutic agent to the fibroid with the needle. Still further, and in addition to or in lieu of energy and/or drug delivery treatments, the fibroid may be treated by mechanical cutting. For example, the fibroid may be morcelated with a tip of the needle. The treatment needle or other elements (e.g., non-treatment needle, thermocouple) may additionally monitor tissue impedance and/or measure a tissue temperature so as to aid in diagnosis, blood supply measurement, thermal signature, tissue targeting, and the like.

(17) In another aspect of the present invention, minimally invasive devices for imaging and treating submucosal, intramural, or subserosal fibroids in one real-time system are provided. The device comprises a sheath, probe, catheter, or other shaft which is adapted for transcervical introduction

into a uterus. A visualization element is within or on a distal steerable portion of the sheath. The visualization element is capable of determining a location of a fibroid on wall of the uterus while the sheath is in the uterus. Typically, the visualization element comprises an ultrasonic transducer. For example, the visualization element may comprise a phased array transducer having 64 elements or a mechanically scanned transducer. Still further, the element may comprise other visualization means, such as hysteroscopy, that is capable of producing a visual image. At least one self-anchoring treatment needle is within or on a distal portion of the sheath. The treatment needle is deployable against the fibroid whose position has been located by the visualization element.

(18) The at least one treatment needle, usually two treatment needles, will be anchored by providing a geometry which inhibits displacement after the needles have been deployed.

Exemplary geometries include non-linear, such as arcuate, helical, such as cork screw, curved, co-axial, everting, and like configurations. For example, the geometry may comprise a pair of converging or diverging needles which when deployed in tissue will remain firmly anchored as a result of the opposed geometry. Such geometries may be conveniently referred to as being "self-anchoring." Such anchoring needles advantageously provide targeted treatment of larger volumes (e.g., larger fibroids) with less damage to non-target tissue. The treatment needle may take on a variety of forms, typically having both extended and retracted configurations, and be made from a variety of materials (e.g., nitinol). For example, the treatment needle may comprise electrodes, electrosurgical needles, or other tissue-penetrating elements capable of delivering ablative radio-frequency energy to target and treat the tumors. Alternatively, the treatment needle may comprise an antenna capable of delivering microwave energy to treat the fibroid. Further, the treatment needle may comprise a hollow tube so as to deliver at least one therapeutic agent to the fibroid. Still further, the treatment needle may comprise a cutting tube so as to morcelate the fibroid.

(19) The visualization element will preferably be located near and/or coupled to the treatment needle so that needle positioning, deployment, and treatment is within a surgeon's field of view. The sheath, visualization element, and/or treatment needle may be integrally formed or comprises separate, modular components that are coupleable to one another. For example, the visualization element may comprise a re-usable ultrasound core that may be positioned within a disposable needle carrying sheath. Further, at least a portion of the sheath, visualization element, and/or treatment needle may be steerable, rotatable, deflectable, flexible, pre-shaped, or pre-formed so as to provide transvaginal access to the uterus for identification and treatment of fibroids. An exemplary interventional deployment and imaging system is described in more detail in co-pending U.S. Provisional Patent Application Ser. No. 60/758,881, filed Jan. 12, 2006, which is assigned to the assignee of the present application and incorporated herein by reference.

(20) A further understanding of the nature and advantages of the present invention will become apparent by reference to the remaining portions of the specification and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following drawings should be read with reference to the detailed description. Like numbers in different drawings refer to like elements. The drawings, which are not necessarily to scale, illustratively depict embodiments of the present invention and are not intended to limit the scope of the invention.

(2) FIGS. 1A through 1F illustrate a first embodiment of the method and device comprising converging ablation needles and on board ultrasound imaging constructed in accordance with the principles of the present invention.

(3) FIGS. 2A through 2D illustrate a second embodiment of the method and device comprising diverging ablation needles and on board ultrasound imaging constructed in accordance with the

principles of the present invention.

(4) FIGS. 3A through 3D illustrate a third embodiment of the method and device comprising telescoping ablation needles and on board ultrasound imaging constructed in accordance with the principles of the present invention.

(5) FIGS. 4A through 4F illustrate a fourth embodiment of the method and device comprising an inflatable balloon which provides treatment and on board ultrasound imaging constructed in accordance with the principles of the present invention.

(6) FIGS. 5A through 5C illustrate a fifth embodiment of the method and device comprising another inflatable balloon which provides treatment and on board ultrasound imaging constructed in accordance with the principles of the present invention.

(7) FIG. 6 illustrates a sixth embodiment of the method and device comprising a mechanical cutting element having a morcelating tip and on board ultrasound imaging constructed in accordance with the principles of the present invention.

(8) FIGS. 7A through 7C illustrate a seventh embodiment of the method and device comprising drug delivery needles and on board ultrasound imaging constructed in accordance with the principles of the present invention.

(9) FIG. 8 illustrates an eighth embodiment of the method and device comprising laproscopically injecting bubbles containing drugs that are activated by intra-uterine ultrasound imaging.

(10) FIGS. 9A and 9B illustrate impedance monitoring for directed fibroid treatment which may be employed with the present invention.

(11) FIG. 10 illustrates a method of laproscopically imaging and treating a fibroid.

(12) FIGS. 11A through 11C illustrate methods of decoupling the ultrasound imaging from the steerable, flexible needle catheter.

(13) FIG. 12 illustrates a direct transuterine diagnostic ultrasound imager.

(14) FIGS. 13A and 13B illustrate schematics of a system constructed in accordance with the principles of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(15) Referring now to FIGS. 1A through 1F, a first embodiment of the invention is illustrated including two converging ablation needles 14 and an ultrasound imaging module 12. A flexible, steerable catheter 10 is shown that acts as a sheath for the ultrasound catheter 12. In FIG. 1A, the two treatment needles 14 are in a retracted configuration within the sheath 10. In FIG. 1B, the ultrasound catheter 12 is shown within the sheath 10 with the two treatment needles 14 in a deployed configuration. One or both converging ablation needles 14 may have insulating sleeves so as to prevent treating non-target tissue and/or thermocouples at a tip region to measure a tissue temperature. FIG. 1C shows application of radiofrequency ablation energy between the two bipolar needle electrodes 14 and the resulting energy field 16 therebetween. FIG. 1D shows the sheath 10 inserted into the uterus 18 via the cervix 20 with a flexible shaft portion 22. As described above, the ultrasound beam 12 not only allows for identification of the fibroids 24, 26, but also serves to provide real-time visualization of needle anchoring and ablation treatment. The ultrasound catheter 12 may further be steered, rotated, or deflected independently of the treatment needles 14 so as to allow for a complete reconstruction view. For example, the ultrasound catheter may be torqued or rotated so that positioning of both needles 14 and treatment 16 may be verified. FIG. 1E show deployment of the treatment needles 14 during ultrasound visualization while FIG. 1F shows radiofrequency ablation treatment 16 of the fibroid tumor 24. Generally, each individual fibroid tumor 24, 26 will be navigated to, imaged, targeted and treated separately. It will be appreciated that the above depictions are for illustrative purposes only and do not necessarily reflect the actual shape, size, or dimensions of the device. This applies to all depictions hereinafter.

(16) Referring now to FIGS. 2A through 2D, a second embodiment of the invention is illustrated including two diverging ablation needles 28 and the ultrasound imaging module 12. Again, the flexible, steerable catheter 10 is shown acting as a sheath for the ultrasound catheter 12. In FIG.

2A, the ultrasound catheter **12** is inserted into the sheath **10** and is visualizing the fibroid tumor **24** within the uterus in an imaging field transverse to an axis of the sheath as denoted by the dashed lines **30**. A hollow nitinol needle **32** is deployed through a lumen **34** in the sheath **10** as illustrated in FIG. 2B. Thereafter, two hooked treatment needles **28** are deployed through the hollow needle **32** within the limits of the imaging field as illustrated in FIG. 2C and anchored against the fibroid **24**. Radiofrequency ablative energy is then delivered in a bipolar fashion between the two poles of the hooked treatment needles **28** while the treatment needles remain within the limits of the imaging field so as to necrose the fibroid tissue **24** as illustrated in FIG. 2D. Fibroid identification, needle deployment, and ablation treatment are carried out under ultrasound visualization **30** in real-time. It will be appreciated that the distances (as denoted by arrows **36**, **38**) that each treatment needle **28** is deployed within the fibroid tissue **24** may be adjusted based on the size of the lesion.

(17) Referring now to FIGS. 3A through 3D, a third embodiment of the invention is illustrated including a telescoping ablation needle **40** and the ultrasound imaging module **12**. Again, the flexible, steerable catheter **10** is shown acting as a sheath for the ultrasound catheter **12**. As shown in FIG. 3A, the ultrasound catheter **12** is inserted into the sheath **10**. The sheath **10** is transcervically introduced into the uterus and used for visualizing the fibroid tumor **24** as denoted by the dashed lines **30**. Similar to FIG. 2B, a first nitinol needle **32** is deployed through the lumen **34** in the sheath **10**. Thereafter, a second telescoping needle **40** is deployed through the first needle **32** as illustrated in FIG. 3C. Radiofrequency ablative energy is delivered in a bipolar fashion between the two telescoping needles **40**, **32** under ultrasound visualization **30**. Again, the distance (as denoted by arrow **42**) that the telescoping treatment needle **40** is extended within the fibroid tissue **24** may be adjustable to the size of the lesion.

(18) Referring now to FIGS. 4A through 4F, a fourth embodiment of the invention is illustrated including an inflatable treatment balloon **44** and the ultrasound imaging catheter **12**. As shown in FIG. 4A, the flexible, steerable sheath **10** is inserted into the uterus **18** via the cervix **20** with the ultrasound module **12** on board. The sheath **10** further has a lumen for insertion of a rotary cutting tube **46**, treatment needle, or other penetrating device. FIG. 4B illustrates visualization of the individual fibroid tumor **24** from within the uterine cavity **18** by the ultrasound module **12**, as denoted by the dashed lines **30**. FIG. 4C illustrates advancement and penetration of the rotary cutting tube **46** into the fibroid tumor **24** under direct visualization **30** through the ultrasound module **12**. The distal end of the rotary cutting tube **46** is depicted with a morcelating tip **47**. In FIG. 4D, some of the fibroid tissue **24** is removed through the rotary cutting tube **46** to create room for the treatment balloon **44**. The rotary cutting tube **46** is further partially retracted to make room in the tumor **24** for the treatment balloon **44**. As shown in FIG. 4E, the treatment balloon **44** is then deployed through the cutting tube **46** and into the tumor **24** under direct visualization **30** through the ultrasound module **12**. As shown in FIG. 4F, the treatment balloon **44** is inflated and ablative energy is applied by the balloon **44** to treat the tumor **24** under direct visualization **30** through the ultrasound module **12**. The ablative energy may comprise any of the energy sources described herein including radiofrequency energy, microwave energy, laser energy, cryo energy, ultrasound energy, HIFU, or radiation. Alternatively or in addition to the treatment balloon **44**, a radiofrequency basket electrode may be disposed over the balloon to treat the tumor.

(19) Referring now to FIGS. 5A through 5C, a fifth embodiment of the invention is illustrated including the inflatable treatment balloon **44** and the ultrasound imaging catheter **12** of FIG. 4F. This embodiment differs in how the treatment balloon **44** is deployed into the tumor **24**. After identification of the fibroid tumor **24** from within the uterine cavity **18**, a rotary cutting tube **48** without a morcelating tip is advanced and penetrated into the fibroid tumor **24** under direct visualization **30** through the ultrasound module **12** as shown in FIG. 5A. A wire **50** is then advanced into the tumor **24** through the cutting tube **48** under direct visualization **30** through the ultrasound module **12** in FIG. 5B. In FIG. 5C, the treatment balloon **44** is advanced through the cutting tube **48** and over the wire **50** and then inflated in the tumor **24** under direct ultrasound

visualization **30** so as to treat the tumor **24** with ablative energy. The ablative energy may comprise any of the energy sources described herein including radiofrequency energy, microwave energy, laser energy, cryo energy, ultrasound energy, HIFU, or radiation.

(20) Referring now to FIG. **6**, a sixth embodiment of the invention is illustrated including the rotary cutting tube **46** and the ultrasound imaging catheter **12** of FIG. **4C**. This embodiment differs in that the rotary cutting tube **46** itself provides treatment of the tumor **24** with its mechanical cutting element having a morcelating tip **47**. After identification of the fibroid tumor **24** and advancement/penetration of the rotary cutting tube **46** into the fibroid tumor **24** under direct visualization **30** through the ultrasound module **12** in the uterus **18**, the fibroid **24** is morcelated or liquefied by the rotary cutting tube **46** and the fibroid tissue **24** is suctioned out through the hollow cutting tube **46** as depicted by reference numeral **52**.

(21) Referring now to FIGS. **7A** through **7C**, a seventh embodiment of the invention is illustrated including a drug delivery needle **54** and the ultrasound imaging catheter **12**. In FIG. **7A**, under ultrasound visualization in the uterus, two treatment needles **54** are anchored within the fibroid **24** and the fibroid treated by the delivery of at least one therapeutic agent **56** to the fibroid with the needles **54**. It will be appreciated that the treatment needles **54** may have both a retracted and extended position and may be adjustable so as to achieve the desired drug delivery profile. Further, drug delivery may take place through a single treatment needle **54** or through multiple treatment needles **54**. The therapeutic agent **56** may comprise a variety of agents. For example, the agent **56** may comprise a chemotherapeutic or chemoablative agent (e.g., alcohol or a chemokine), a gene therapy agent, a tissue necrosis agent, an antibody, or the like. The drug delivery needles **54** may treat tumors of various sizes. For example, FIG. **7B** illustrates treatment of a large tumor **24'** (e.g., 40 mm), while FIG. **7C** illustrates treatment of a smaller tumor **24''** (e.g., 20 mm).

(22) Referring now to FIG. **8**, another drug delivery method and device is illustrated. A syringe **58** is used to laproscopically inject contrast bubbles **60** containing at least one therapeutic agent **56** into the fibroid **24** instead of transcervical drug delivery via treatment needles **54**. After drug delivery injection into the fibroid **24**, the ultrasound imaging catheter **12** in the uterus **18** activates the agent **56** by targeted ultrasound **30**. For example, this may cause the bubbles **60** to burst or break in the fibroid blood supply **24** which in turn releases the therapeutic agent **56** to the fibroid **24** for treatment.

(23) Referring now to FIG. **9A**, the flexible, steerable catheter **10** is shown inserted into the uterus **18** via the cervix **20**. The catheter **10** has an on board ultrasound imaging module **12** and a lumen for insertion of at least one needle **62** or other penetrating device. In this illustration, multiple needles **62** are shown inserted into the fibroid tumor **24** with impedance monitoring to denote the change in the impedance of the tissue from inside the tumor **24** versus tissue outside the tumor **24** and/or tissue outside the uterine wall. Impedance monitoring will aid in directly targeting the fibroid tumor **24** for treatment (e.g., energy delivery, drug delivery, mechanical cutting, etc.) and may also safely control treatment delivery so that it is only within the uterus **18** itself. Further, impedance profiling may denote border recognition of tissue. This in turn may allow for implementation of additional safety mechanisms. For example, automatic shutoff of the device may be implemented if the needle **62** is extended beyond the fibroid **24** and/or uterus **18**.

(24) Referring now to FIG. **9B**, the flexible, steerable catheter **10** is shown inserted uterus **18** via the cervix **20**. The catheter **10** has an on board ultrasound imaging module **12** and a lumen for insertion of at least one needle **64** or other penetrating device. The needle **64** is shown inserted into the fibroid tumor **24** with impedance monitoring to denote the change in the impedance of the tissue from inside the tumor **24** versus tissue outside the tumor and/or tissue outside the uterine wall. Impedance monitoring will aid in directly targeting the fibroid tumor **24** for treatment from the uterine wall.

(25) Referring now to FIG. **10**, a flexible, steerable laparoscopic probe **10** is shown accessing the uterus **18** from an abdominal port **66** in the abdominal wall **68**. The probe **10** uses the ultrasound

module **12** outside of the uterus **18** to target fibroid tumors **24** that are within the uterus **18**. The probe **10** then uses the treatment needle **70** under direct visualization **30** through the ultrasound module **12** to then treat the fibroid **24** with ablative energy.

(26) Referring now to FIG. **11A**, a flexible, steerable catheter based probe **10** having a treatment needle **72** is shown inserted into the uterus **18** and within the fibroid **24** using a non-coupled vaginal ultrasound probe **74**. The two devices **10**, **74** are operated independently of each other. Referring now to FIG. **11B**, the flexible, steerable needle catheter **10** is shown inserted into the uterus **18** and the treatment needle **72** within the fibroid **24** using a non-coupled abdominal ultrasound probe **76**. The two devices **10**, **76** are operated independently of each other. With respect to FIG. **11C**, the flexible, steerable laparoscopic needle probe **10** is shown accessing the uterus **18** from an abdominal port **66** in the abdominal wall **68**. The treatment needle **70** of the probe **10** is shown accessing the fibroid **24** with the aid of ultrasound visualization **30** from the abdominal ultrasound probe **76**. The two devices **70**, **76** are operated independently of each other.

(27) Referring now to FIG. **12**, a flexible, steerable intrauterine ultrasound imaging device **78** is shown for imaging the uterine wall and lining transendometrially for the diagnosis of fibroids **24**, **26**. The ultrasound imaging head **82** generally comprises an ultrasonic phased array transducer having 64 elements. The ultrasound transducer may also be mechanical, linear, or curved. A sterile drape **80** may be placed over the diagnostic imager **78**, wherein a gel may be used within the drape **80** for improved image coupling. The diagnostic imager **78** may also be used without a drape **80**, when disposable, using natural body fluids for image coupling. The diagnostic imager **78** further has a flexible section **84** capable of deflection in a range from 0 degrees to about 90 degrees via an angle adjustment knob **86**. The diagnostic ultrasound imager **78** is inserted directly into the uterine cavity **18**, either with or without dilation of the cervix **20**, in order to directly image the fibroids **24**, **26** within the wall of the uterus **18**. This imaging provides a closer and more direct view of the tumors **24**, **46** in order to more accurately diagnose the location and characterization of the fibroids or other pathology.

(28) FIGS. **13A** and **13B** illustrate schematics of a system constructed in accordance with the principles of the present invention. The system comprises a combined ultrasound recognition and radiofrequency treatment system **88**. The system **88** may provide a variety of features including ultrasound mapping, ultrasound recognition of treatment area (e.g., tissue differentiation via temperature profiling), radiofrequency ablation treatment under ultrasound imaging, temperature monitoring, time monitoring, and/or impedance monitoring. The system **88** may be coupled to various devices **90** described herein having single or multiple treatment needle configurations to ablate in either bipolar or monopolar modes.

(29) Although certain exemplary embodiments and methods have been described in some detail, for clarity of understanding and by way of example, it will be apparent from the foregoing disclosure to those skilled in the art that variations, modifications, changes, and adaptations of such embodiments and methods may be made without departing from the true spirit and scope of the invention. Therefore, the above description should not be taken as limiting the scope of the invention which is defined by the appended claims.

Claims

1. A device for treating a uterine fibroid, the device comprising: a probe; a visualization element disposed on the probe and configured to take a real-time image of the fibroid from inside a uterus when the probe is advanced therein; a plurality of needles configured to deploy into the fibroid while the real-time image of the fibroid being taken using the visualization element is being displayed, wherein the plurality of needles are configured to extend away from a location on the probe in a converging or diverging manner into the fibroid, and wherein the plurality of needles are configured to treat the fibroid; and wherein the visualization element is moveable independent of

the plurality of needles.

2. The device of claim 1, wherein the plurality of needles is configured to treat the fibroid by delivering ablative energy.

3. The device of claim 2, wherein the ablative energy is one or more of electrical, radiofrequency (RF), laser, microwave, freezing or cryogenic, ultrasound, high frequency ultrasound (HIFU), or radiative.

4. The device of claim 1, wherein the visualization element comprises an ultrasound transducer.

5. The device of claim 4, wherein the ultrasonic transducer comprises a phased array transducer having 64 elements or a mechanically scanned transducer.

6. The device of claim 1, wherein one or more needles of the plurality of needles has an arcuate geometry.

7. The device of claim 1, wherein one or more needles of the plurality of needles comprise an electrode, antenna, hollow tube, or cutting tube.

8. The device of claim 1, wherein one or more needles of the plurality of needles is formed from Nitinol.

9. The device of claim 1, wherein the plurality of needles is configured to treat the fibroid by delivering at least one therapeutic agent to the fibroid.

10. The device of claim 1, wherein the visualization element is one or more of steerable, rotatable, deflectable, flexible, pre-shaped, or pre-formed.

11. The device of claim 1, wherein the plurality of needles comprises at least two treatment needles.

12. The device of claim 1, wherein one or more needles of the plurality of needles has one or more of a non-linear, straight, linear, arcuate, helical, cork screw, or curved geometry when deployed.

13. The device of claim 1, wherein one or more needles of the plurality of needles is configured to measure tissue impedance.

14. The device of claim 1, wherein the plurality of needles comprises a first needle and a plurality of second needles, and wherein at least one needle of the plurality of second needles is configured to extend out from an open distal end of the first needle.

15. The device of claim 14, wherein one or more of (i) the first needle or (ii) the plurality of second needles are deployable from one or more locations on the probe proximal of the visualization element.

16. The device of claim 15, wherein the first needle and the plurality of second needles are deployable from locations proximal of the visualization element to converge or diverge within the tissue of the fibroid within the field of view of the visualization element.

17. The device of claim 1, wherein the probe, visualization element, and plurality of needles are integrally formed.

18. The device of claim 1, wherein the probe, visualization element, and plurality of needles comprise separate, modular components that are configured to be coupled to one another.

19. The device of claim 1, wherein one or more of (i) at least a portion of the probe, (ii) the visualization element, or (iii) the plurality of needles are one or more of steerable, rotatable, deflectable, flexible, pre-shaped, or pre-formed.

20. The device of claim 1, wherein the device is configured to be minimally invasive.
